

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

THINA RAJKUMAR R
25ITR161

JAVA ASSIGNMENT

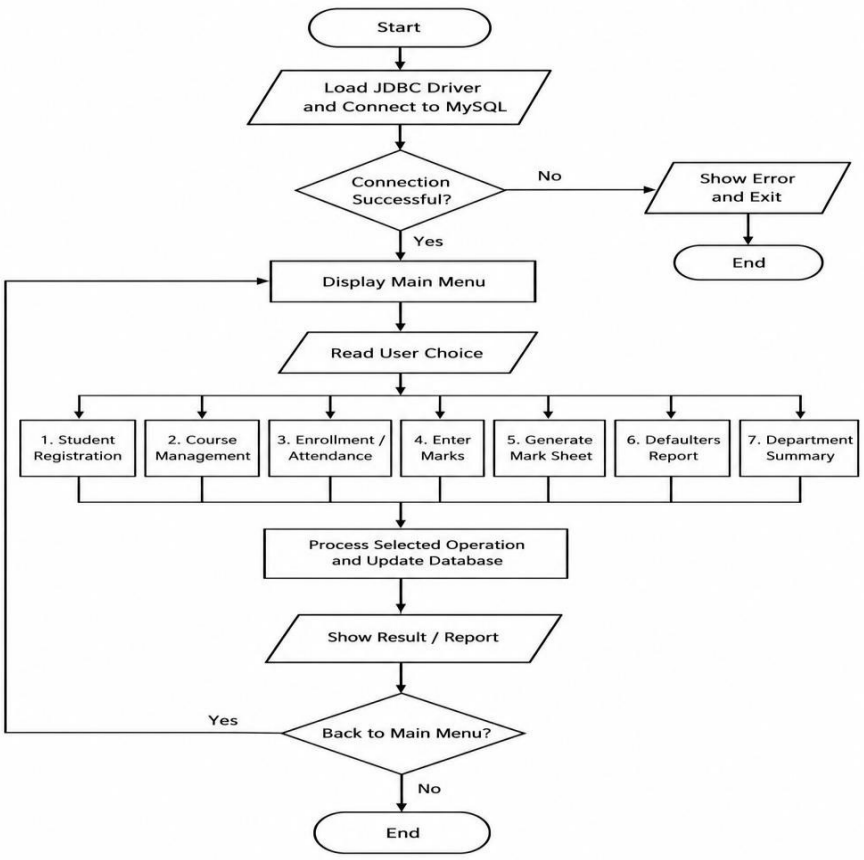
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector Java driver to establish seamless communication between the Java application and the database.

- Establish a connection to the local MySQL database using the correct login details.

- Add and manage student information, including roll number, name, department, and semester, along with course details such as code, name, credits, and category.

- Register students for courses and record their attendance by maintaining the number of classes held and attended.

- Decide exam eligibility based on attendance percentage. Students with 75% or above can attend exams, those with 65% to 74% need medical condonation, and those below 65% are marked as attendance defaulters.

- Enter and maintain internal assessment marks for tests, assignments, quizzes, and practical work.

- Record end-semester exam marks and identify absent students. Ensure that students meet the minimum passing marks for each course type.

- Calculate total marks using internal and external scores, assign grades and grade points, and identify students who need to reappear for exams.

- Prepare semester mark sheets with internal marks, external marks, total marks, and GPA. Also, generate attendance defaulter lists and show the pass percentage for each department through the console menu

DATABASE SCHEMA:

1.STUDENTS TABLE:

```
1.STUDENTS TABLE:
mysql> describe students;
+------------------+--------------+------+-----+-------------------+-------------------+
| Field            | Type         | Null | Key | Default           | Extra             |
+------------------+--------------+------+-----+-------------------+-------------------+
| roll_no          | varchar(20)  | NO   | PRI | NULL              |                   |
| name             | varchar(100) | NO   |     | NULL              |                   |
| department       | varchar(20)  | NO   |     | NULL              |                   |
| current_semester | int          | NO   |     | 3                 |                   |
| created_at       | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.03 sec)
```

2. COURSES TABLE:

```
2. COURSES TABLE:
mysql> describe courses;
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                              | Null | Key | Default           | Extra             |
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
| course_code | varchar(20)                       | NO   | PRI | NULL              |                   |
| title       | varchar(150)                      | NO   |     | NULL              |                   |
| credits     | int                               | NO   |     | NULL              |                   |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED')| NO   |     | NULL              |                   |
| semester    | int                               | NO   |     | 3                 |                   |
| created_at  | timestamp                         | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.01 sec)
```

3. ENROLLMENTS TABLE:

```
3. ENROLLMENTS TABLE:
mysql> describe enrollments;
+-----------------+-------------+------+-----+-------------------+-------------------+
| Field           | Type        | Null | Key | Default           | Extra             |
+-----------------+-------------+------+-----+-------------------+-------------------+
| enrollment_id   | int         | NO   | PRI | NULL              | auto_increment    |
| roll_no         | varchar(20) | NO   | MUL | NULL              |                   |
| course_code     | varchar(20) | NO   | MUL | NULL              |                   |
| semester        | int         | NO   |     | 3                 |                   |
| total_classes   | int         | NO   |     | 0                 |                   |
| attended_classes| int         | NO   |     | 0                 |                   |
| is_condoned     | tinyint(1)  | NO   |     | 0                 |                   |
| created_at      | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----------------+-------------+------+-----+-------------------+-------------------+
8 rows in set (0.01 sec)
```

4. ASSESSMENT MARKS TABLE:

```
4. ASSESSMENT MARKS TABLE:
mysql> describe assessment_marks;
+------------------+--------------+------+-----+-------------------+-----------------------------------------------+
| Field            | Type         | Null | Key | Default           | Extra                                         |
+------------------+--------------+------+-----+-------------------+-----------------------------------------------+
| mark_id          | int          | NO   | PRI | NULL              | auto_increment                                |
| enrollment_id    | int          | NO   | UNI | NULL              |                                               |
| cia1_marks       | decimal(5,2) | YES  |     | NULL              |                                               |
| cia2_marks       | decimal(5,2) | YES  |     | NULL              |                                               |
| assignment_marks | decimal(5,2) | YES  |     | NULL              |                                               |
| ese_marks        | decimal(5,2) | YES  |     | NULL              |                                               |
| is_ese_absent    | tinyint(1)   | NO   |     | 0                 |                                               |
| updated_at       | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+------------------+--------------+------+-----+-------------------+-----------------------------------------------+
8 rows in set (0.02 sec)

C:\Users\Thina rajkumar\Downloads\KEC_Student_Evaluation_System>
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.util.*;


public class MainApp {
    private static final AcademicDAO dao = new AcademicDAO();
    private static final Scanner sc = new Scanner(System.in);


    public static void main(String[] args) {
        if (!DBConnection.testConnection()) { System.out.println("DB Connection failed."); return; }
        System.out.println("=== KEC EVALUATION SYSTEM (R2024) ===");
        boolean run = true;
        while (run) {
            System.out.println("\n1.Student 2.Course 3.Enroll/Attn 4.Marks 5.Marksheet 6.Defaulters 7.Summary 0.Exit");
            System.out.print("Choice: ");
            switch (sc.nextLine().trim()) {
                case "1": regStudent(); break;
                case "2": addCourse(); break;
```

```
            case "3": enrollStudent(); break;
            case "4": enterMarks(); break;
            case "5": System.out.print("Roll: "); dao.printStudentMarkSheet(sc.nextLine().trim(), in("Sem: ")); break;
            case "6": dao.printAttendanceDefaulters(in("Sem: ")); break;
            case "7": dao.printDepartmentSummary(in("Sem: ")); break;
            case "0": run = false; break;
            default: System.out.println("Invalid choice.");
        }
    }
    System.out.println("System closed.");
  }
    while (run) {
        System.out.println("\n1.Student 2.Course 3.Enroll/Attn 4.Marks 5.Marksheet 6.Defaulters 7.Summary 0.Exit");
        System.out.print("Choice: ");
        switch (sc.nextLine().trim()) {
            case "1": regStudent(); break;
            case "2": addCourse(); break;
            case "3": enrollStudent(); break;
            case "4": enterMarks(); break;
            case "5": System.out.print("Roll: "); dao.printStudentMarkSheet(sc.nextLine().trim(), in("Sem: ")); break;
            case "6": dao.printAttendanceDefaulters(in("Sem: ")); break;
            case "7": dao.printDepartmentSummary(in("Sem: ")); break;
```

```
            case "0": run = false; break;
            default: System.out.println("Invalid choice.");
        }
    }
    System.out.println("System closed.");
  }
  private static void regStudent() {
    System.out.println(dao.registerStudent(str("Roll: "), str("Name: "), str("Dept: "), in("Sem: ")) ? "Registered." : "Failed.");
  }
  private static void addCourse() {
    System.out.println(dao.addCourse(str("Code: "), str("Title: "), in("Credits: "), str("Type (THEORY/PRACTICAL/EMBEDDED): ")) ? "Added." : "Failed.");
  }
  private static void enrollStudent() {
    String ch = str("1.Enroll 2.Attn: "), r = str("Roll: "), c = str("Course: "); int s = in("Sem: ");
    if (ch.equals("1")) {
      System.out.println(dao.enrollStudent(r, c, s) > 0 ? "Enrolled." : "Failed.");
    } else {
      int id = dao.getEnrollmentId(r, c, s);
      if (id <= 0) { System.out.println("Not found."); return; }
      System.out.println(dao.updateAttendance(id, in("Total: "), in("Attended: "), str("Condonation (y/n): ").equalsIgnoreCase("y")) ? "Updated." : "Failed.");
    }
```

```
    }
    private static void enterMarks() {
        int id = dao.getEnrollmentId(str("Roll:  "), str("Course: "), in("Sem: "));
        if (id <= 0) { System.out.println("Not  found."); return; }
        String p = str("1.CIA 2.Assignment 3.ESE: ");
        if (p.equals("1")) System.out.println(dao.recordPhase1Marks(id,  db("CIA-1: "), db("CIA-
2: ")) ? "Saved." : "Failed.");
        else  if (p.equals("2"))  System.out.println(dao.recordPhase2Marks(id,   db("Assignment:
")) ? "Saved." : "Failed.");
        else if (p.equals("3")) {
            boolean abs = str("Absent? (y/n):  ").equalsIgnoreCase("y");
            System.out.println(dao.recordPhase3Marks(id,  abs ? null : db("ESE: "), abs) ? "Saved."
: "Failed.");
        }
    }
    private static String str(String p) { System.out.print(p);  return sc.nextLine().trim();  }
    private static int in(String p) { return Integer.parseInt(str(p));  }
    private static double db(String p) { return Double.parseDouble(str(p)); }
}
```

DB CONNECTION.JAVA:

```
import java.sql.Connection; import
java.sql.DriverManager;
import java.sql.SQLException;
/**
 * DBConnection - Database Connection Management for KEC Evaluation System
 * Kongu Engineering College (Autonomous) - Regulation 2024
 */
public class DBConnection {
    private static final String DEFAULT_HOST = "localhost";
    private static final String DEFAULT_PORT = "3306";
    private static final String DEFAULT_DB   = "kec_eval";
    private static final String DEFAULT_USER = "root";
    private static final String DEFAULT_PASS = "0000";

    public static String getUrl() {
        String host = System.getenv("DB_HOST") != null ? System.getenv("DB_HOST") :
DEFAULT_HOST;
        String port = System.getenv("DB_PORT") != null ? System.getenv("DB_PORT") :
DEFAULT_PORT;
        String db   = System.getenv("DB_NAME") != null ? System.getenv("DB_NAME") :
DEFAULT_DB;
        return "jdbc:mysql://" + host + ":" + port + "/" + db +
"?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";
    }

 public static String getUser() {
```

```
        return System.getenv("DB_USER") != null ? System.getenv("DB_USER") : DEFAULT_USER;
    }
    public static String getPassword() {
        return System.getenv("DB_PASSWORD") != null ? System.getenv("DB_PASSWORD") : DEFAULT_PASS;
    }

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("[ERROR] MySQL JDBC Driver not found in classpath: " + e.getMessage());
        }
    }

    /**
     * Obtains a new database connection
     */
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(getUrl(), getUser(), getPassword());
    }

    /**
```

```
     * Tests the database connectivity
     */
    public static boolean testConnection() {
        try (Connection conn = getConnection()) {
            return conn != null && !conn.isClosed();
        } catch (SQLException e) {
            System.err.println("[ERROR] Connection test failed: " + e.getMessage());
            return false;
        }
    }
}
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
================================================================================
              KONGU ENGINEERING COLLEGE (AUTONOMOUS) - ERODE - 638060
           REGULATION 2024 (R2024) - STUDENT ACADEMIC & EVALUATION SYSTEM
               Database-Driven Java JDBC Application (MySQL / SQLite)
================================================================================
--------------------------------------------------------------------------------
                                    MAIN MENU
--------------------------------------------------------------------------------
  1. Student Registration (Add New Student)
  2. Course Catalog Management (Add New Course - Theory/Practical/Embedded)
  3. Course Enrollment & Attendance Tracking (Record Attendance & Condonation)
  4. Incremental Marks Entry (Phase 1: CIA1/2 | Phase 2: Assg/Lab | Phase 3: ESE/AB)
  5. Generate Student Semester Mark Sheet (Detailed with SGPA & %)
  6. Generate Attendance Defaulter Report (Attendance < 65% Barred SA)
  7. Generate Class Performance & Result Summary (Department-wise Pass/Fail)
  8. View All Students, Courses & Active Enrollments
  9. Run Full Automated System Demonstration (Self-Verification)
  0. Exit System
--------------------------------------------------------------------------------
```

STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
Enter your choice (0-8): 1

>>> 1. STUDENT REGISTRATION <<<
Enter Student Roll Number (e.g., 24ITR001): 25ITR161
Enter Student Full Name: Thina Rajkumar R
Enter Department (e.g., Information Technology / CSE / ECE / MECH): IT
Enter Current Semester (1-8): 3
[SUCCESS] Student successfully registered:
         25ITR161     | Thina Rajkumar R          | IT                              | Sem 3
```

2. COURSE CATALOG:

```
Enter your choice (0-8): 2

>>> 2. COURSE CATALOG MANAGEMENT <<<
Enter Course Code (e.g., 24ITT31): 24ITC31
Enter Course Title: Full Stack Development
Enter Credits (1-6): 3
Select Course Type according to KEC R2024:
  1. THEORY     (CIA: 40% | ESE: 60% | Min ESE: 45% = 27/60)
  2. PRACTICAL  (CIA: 60% | ESE: 40% | Min ESE: 50% = 20/40)
  3. EMBEDDED   (CIA: 50% | ESE: 50% | Min ESE: 45% = 22.5/50)
Choice (1-3): 3
[SUCCESS] Course registered successfully in catalog:
          24ITC31    | Full Stack Development                  | 3 Credits | EMBEDDED
```

3. COURSE ENROLLMENT - ENROLL STUDENT IN COURSE:

```
Enter your choice (0-8): 3

>>> 3. COURSE ENROLLMENT & ATTENDANCE TRACKING <<<
  a. Enroll Student in Course
  b. Update Class Attendance (Held vs Attended)
  c. Grant / Revoke Medical Condonation (65% to 74.9%)
Select sub-action (a/b/c): a
Enter Student Roll Number: 25ITR161
Enter Course Code: 24ITC31
Enter Semester: 3
[SUCCESS] Student 25ITR161 enrolled in 24ITC31 (Enrollment ID: 17)
```

ATTENDANCE TRACKING - UPDATE CLASS ATTENDANCE:

```
Enter your choice (0-8): 3

>>> 3. COURSE ENROLLMENT & ATTENDANCE TRACKING <<<
  a. Enroll Student in Course
  b. Update Class Attendance (Held vs Attended)
  c. Grant / Revoke Medical Condonation (65% to 74.9%)
Select sub-action (a/b/c): b
Enter Enrollment ID to update: 17
Current Record: 25ITR161 | 24ITC31 (Full Stack Development) | Held: 0, Attended: 0 (100.00%)
Enter Total Classes Held: 30
Enter Total Classes Attended: 25
[SUCCESS] Attendance updated! New Attendance: 83.33%
[ELIGIBLE] Attendance >= 75%: Fully eligible for End Semester Examination.
```

INCREMENTAL MARKS ENTRY - PHASE 1 (MID-SEMESTER):

```
Enter your choice (0-8): 4

>>> 4. INCREMENTAL MARKS ENTRY (KEC R2024) <<<
The system supports phased incremental assessment entry as the semester progresses:
  1. Phase 1 (Mid-Semester)       : Record CIA Test 1 (0-100) and CIA Test 2 (0-100)
  2. Phase 2 (End of Semester-CIA): Record Assignments / Quizzes / Lab Evaluations (0-100)
  3. Phase 3 (Post Semester)      : Record End Semester Exam (ESE) Marks (0-100) / Absence (AB)
Select Assessment Phase (1-3): 2
Enter Enrollment ID: 17
Target: 25ITR161 (Thina Rajkumar R) | Course: 24ITC31 (Full Stack Development - EMBEDDED)
--- Phase 2: End-of-Semester Continuous Assessments ---
Enter Assignments / Quizzes Marks (0 - 100): 80
[SUCCESS] Phase 2 Recorded: Assignments / Quizzes = 80.0
```

INCREMENTAL MARKS ENTRY - PHASE 2 (END OF SEM CIA) :

```
Enter your choice (0-8): 4

>>> 4. INCREMENTAL MARKS ENTRY (KEC R2024) <<<
The system supports phased incremental assessment entry as the semester progresses:
  1. Phase 1 (Mid-Semester)       : Record CIA Test 1 (0-100) and CIA Test 2 (0-100)
  2. Phase 2 (End of Semester-CIA): Record Assignments / Quizzes / Lab Evaluations (0-100)
  3. Phase 3 (Post Semester)      : Record End Semester Exam (ESE) Marks (0-100) / Absence (AB)
Select Assessment Phase (1-3): 3
Enter Enrollment ID: 17
Target: 25ITR161 (Thina Rajkumar R) | Course: 24ITC31 (Full Stack Development - EMBEDDED)
--- Phase 3: Post-Semester End Semester Examination (ESE) ---
Is student ABSENT in ESE? (y/n): y
[SUCCESS] Student marked as ABSENT (AB) for ESE. Grade awarded: AB (0 Grade Points).
```

INCREMENTAL MARKS ENTRY- PHASE 3 (POST SEMESTER - ESE):

```
Enter your choice (0-8): 6

================================================================================================================
                    KEC R2024 - ATTENDANCE DEFAULTER & ELIGIBILITY REPORT
================================================================================================================
[1] STUDENTS BARRED FROM ESE (Attendance < 65% or Unapproved Condonation - Grade SA Awarded)
----------------------------------------------------------------------------------------------------------------
Roll No    | Student Name         | Course    | Course Title              | Attn% |   Status | Penalty / Action
----------------------------------------------------------------------------------------------------------------
24ITR003   | Vignesh S            | 24ITT11   | Problem Solving & Pyth... | 53.3% | DEF_SA   | Barred (SA - 0 GP)

[2] MEDICAL CONDONATION CASES (65.0% to 74.9% Attendance)
----------------------------------------------------------------------------------------------------------------
Roll No    | Student Name         | Course    | Course Title              | Attn% | Condonation  | ESE Eligibility
----------------------------------------------------------------------------------------------------------------
24ITR002   | Priyadharshini K     | 24ITP11   | Object Oriented Progra... | 70.7% | APPROVED     | ELIGIBLE
24ITR002   | Priyadharshini K     | 24ITT11   | Problem Solving & Pyth... | 70.0% | APPROVED     | ELIGIBLE
================================================================================================================
```

SEMESTER GRADE & PERFORMANCE REPORT (STUDENT MARK SHEET):

```
>>> 8. VIEW CATALOG & ENROLLMENTS <<<
  a. View All Students
  b. View All Courses in Catalog
  c. View All Student Enrollments
Select option (a/b/c): a
Roll Number  | Name                    | Department                        |  Sem
-------------------------------------------------------------------------------
24CSR015     | Ananya R                | Computer Science and Engineering  |    1
24ITR001     | Thina Rajkumar          | Information Technology            |    1
24ITR002     | Priyadharshini K        | Information Technology            |    1
24ITR003     | Vignesh S               | Information Technology            |    1
24ITR099     | Naveen Kumar M          | Information Technology            |    1
24MEC007     | Karthik V               | Mechanical Engineering            |    1
25ITR161     | Thina Rajkumar R        | IT                                |    3
```

VIEW CATALOG & ENROLLMENTS - VIEW ALL STUDENTS :

```
>>> 8. VIEW CATALOG & ENROLLMENTS <<<
  a. View All Students
  b. View All Courses in Catalog
  c. View All Student Enrollments
Select option (a/b/c): b
Code       | Title                                      | Credits | Type
-------------------------------------------------------------------------------
24ITC31    | Full Stack Development                     |       3 | EMBEDDED
24ITL11    | Programming Practices Laboratory           |       2 | PRACTICAL
24ITP11    | Object Oriented Programming with Java      |       4 | EMBEDDED
24ITT11    | Problem Solving & Python Programming       |       3 | THEORY
24ITT12    | Computer Organization & Architecture       |       3 | THEORY
24MAT11    | Matrices and Calculus                      |       4 | THEORY
```

VIEW CATALOG & ENROLLMENTS – VIEW ALL COURSES IN CATALOG:

```
>>> 8. VIEW CATALOG & ENROLLMENTS <<<
  a. View All Students
  b. View All Courses in Catalog
  c. View All Student Enrollments
Select option (a/b/c): c
ID   | Roll No    | Student Name        | Code     | Course Title             | Type     | Attn %
-------------------------------------------------------------------------------------------
13   | 24CSR015   | Ananya R            | 24ITT11  | Problem Solving & Pyth... | THEORY   |  91.7%
14   | 24CSR015   | Ananya R            | 24MAT11  | Matrices and Calculus     | THEORY   |  86.7%
3    | 24ITR001   | Thina Rajkumar      | 24ITL11  | Programming Practices ... | PRACTICAL |  97.8%
2    | 24ITR001   | Thina Rajkumar      | 24ITP11  | Object Oriented Progra... | EMBEDDED |  96.0%
1    | 24ITR001   | Thina Rajkumar      | 24ITT11  | Problem Solving & Pyth... | THEORY   |  93.3%
4    | 24ITR001   | Thina Rajkumar      | 24MAT11  | Matrices and Calculus     | THEORY   |  95.0%
7    | 24ITR002   | Priyadharshini K    | 24ITL11  | Programming Practices ... | PRACTICAL |  75.6%
6    | 24ITR002   | Priyadharshini K    | 24ITP11  | Object Oriented Progra... | EMBEDDED |  70.7%
5    | 24ITR002   | Priyadharshini K    | 24ITT11  | Problem Solving & Pyth... | THEORY   |  70.0%
8    | 24ITR002   | Priyadharshini K    | 24MAT11  | Matrices and Calculus     | THEORY   |  76.7%
11   | 24ITR003   | Vignesh S           | 24ITL11  | Programming Practices ... | PRACTICAL |  84.4%
10   | 24ITR003   | Vignesh S           | 24ITP11  | Object Oriented Progra... | EMBEDDED |  82.7%
9    | 24ITR003   | Vignesh S           | 24ITT11  | Problem Solving & Pyth... | THEORY   |  53.3%
12   | 24ITR003   | Vignesh S           | 24MAT11  | Matrices and Calculus     | THEORY   |  80.0%
16   | 24ITR099   | Naveen Kumar M      | 24ITT12  | Computer Organization ... | THEORY   |  88.0%
15   | 24MEC007   | Karthik V           | 24ITT11  | Problem Solving & Pyth... | THEORY   |  83.3%
17   | 25ITR161   | Thina Rajkumar R    | 24ITC31  | Full Stack Development    | EMBEDDED |  83.3%
```

VIEW CATALOG & ENROLLMENTS -VIEW ALL STUDENT ENROLLMENTS:

```
Enter your choice (0-8): 3

>>> 3. COURSE ENROLLMENT & ATTENDANCE TRACKING <<<
  a. Enroll Student in Course
  b. Update Class Attendance (Held vs Attended)
  c. Grant / Revoke Medical Condonation (65% to 74.9%)
Select sub-action (a/b/c): b
Enter Enrollment ID to update: 17
Current Record: 25ITR161 | 24ITC31 (Full Stack Development) | Held: 0, Attended: 0 (100.00%)
Enter Total Classes Held: 30
Enter Total Classes Attended: 25
[SUCCESS] Attendance updated! New Attendance: 83.33%
[ELIGIBLE] Attendance >= 75%: Fully eligible for End Semester Examination.
```

EXIT SYSTEM - THANK YOU MESSAGE:

```
================================================================================================================
                        KONGU ENGINEERING COLLEGE (AUTONOMOUS) - ERODE
                              SEMESTER GRADE & PERFORMANCE REPORT
                                 REGULATION 2024 (R2024)
================================================================================================================
  Roll Number : 25ITR161          Student Name : Thina Rajkumar R
  Department  : IT                Semester     : 3
----------------------------------------------------------------------------------------------------------------
Code      | Course Title                | Type      | Crd | Attn % |   CIA |   ESE |   Tot |   Grd | GP | Result
----------------------------------------------------------------------------------------------------------------
24ITC31   | Full Stack Development      | EMBEDDED  |   3 |  83.3% |  41.8 |   AB  |  41.8 |  AB |  0 | ABSENT (AB)
================================================================================================================
  Total Enrolled Credits  : 3           Total Credits Earned (Passed) : 0
  Semester SGPA (0 - 10)  : 0.00        Equivalent Percentage (%)    : 0.00 %
================================================================================================================
  Weightage Rules: THEORY (40% CIA / 60% ESE) | PRACTICAL (60% CIA / 40% ESE) | EMBEDDED (50% / 50%)
  Grading Scale  : O (91-100: 10), A+ (81-90: 9), A (71-80: 8), B+ (61-70: 7), B (50-60: 6), RA/SA/AB: 0
================================================================================================================
```

SEMESTER GRADE & PERFORMANCE REPORT:

```
Enter your choice (0-8): 6

================================================================================================================
                     KEC R2024 - ATTENDANCE DEFAULTER & ELIGIBILITY REPORT
================================================================================================================
[1] STUDENTS BARRED FROM ESE (Attendance < 65% or Unapproved Condonation - Grade SA Awarded)
----------------------------------------------------------------------------------------------------------------
Roll No    | Student Name         | Course    | Course Title              | Attn% |   Status | Penalty / Action
----------------------------------------------------------------------------------------------------------------
24ITR003   | Vignesh S            | 24ITT11   | Problem Solving & Pyth... | 53.3% | DEF_SA   | Barred (SA - 0 GP)

[2] MEDICAL CONDONATION CASES (65.0% to 74.9% Attendance)
----------------------------------------------------------------------------------------------------------------
Roll No    | Student Name         | Course    | Course Title              | Attn% | Condonation  | ESE Eligibility
----------------------------------------------------------------------------------------------------------------
24ITR002   | Priyadharshini K     | 24ITP11   | Object Oriented Progra... | 70.7% | APPROVED     | ELIGIBLE
24ITR002   | Priyadharshini K     | 24ITT11   | Problem Solving & Pyth... | 70.0% | APPROVED     | ELIGIBLE
================================================================================================================
```

DATABASE:

STUDENTS DATABASE:

```
Enter your choice (0-8): 0

Thank you for using KEC R2024 Academic Evaluation System.

Press any key to continue . . .
```

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
```

course_code	title	credits	course_type	semester	created_at
24ADC31	Artificial Intelligence & Knowledge Rep.	4	EMBEDDED	3	2026-10-19 20:59:10
24ADC41	Machine Learning Algorithms & Systems	4	EMBEDDED	4	2026-10-19 20:59:10
24ADC51	Natural Language Processing (NLP)	4	EMBEDDED	5	2026-10-19 20:59:10
24ADC61	Generative AI & Large Language Models	4	EMBEDDED	6	2026-10-19 20:59:10
24ADC71	Reinforcement Learning & Autonomous Agents	4	EMBEDDED	7	2026-10-19 20:59:10
24ADL31	Python for AI & Data Science Laboratory	1	PRACTICAL	3	2026-10-19 20:59:10
24ADL41	Machine Learning & Data Mining Lab	1	PRACTICAL	4	2026-10-19 20:59:10
24ADL51	Deep Learning & NLP Laboratory	1	PRACTICAL	5	2026-10-19 20:59:10
24ADL61	Computer Vision & GenAI Laboratory	1	PRACTICAL	6	2026-10-19 20:59:10
24ADP71	AI Capstone Project Phase I	3	PRACTICAL	7	2026-10-19 20:59:10
24ADP81	AI Capstone Project Phase II / Industry Track	10	PRACTICAL	8	2026-10-19 20:59:10
24ADT31	Mathematical Foundations for Data Science	4	THEORY	3	2026-10-19 20:59:10
24ADT41	Statistical Learning & Data Mining	4	THEORY	4	2026-10-19 20:59:10
24ADT51	Deep Learning & Neural Networks	4	THEORY	5	2026-10-19 20:59:10
24ADT61	Computer Vision & Video Analytics	4	THEORY	6	2026-10-19 20:59:10
24ADT71	AI Ethics, Governance & MLOps Infrastructure	3	THEORY	7	2026-10-19 20:59:10
24CEC31	Surveying and Geomatics	4	EMBEDDED	3	2026-10-19 20:59:10
24CEC41	Fluid Mechanics & Hydraulic Machinery	4	EMBEDDED	4	2026-10-19 20:59:10
24CEC51	Soil Mechanics & Geotechnical Engineering	4	EMBEDDED	5	2026-10-19 20:59:10
24CEC61	Environmental Engineering & Water Treatment	4	EMBEDDED	6	2026-10-19 20:59:10
24CEC71	Transportation Engineering & GIS	4	EMBEDDED	7	2026-10-19 20:59:10
24CEL31	Surveying Field Work Laboratory	1	PRACTICAL	3	2026-10-19 20:59:10
24CEL41	Materials Testing & Hydraulics Lab	1	PRACTICAL	4	2026-10-19 20:59:10
24CEL51	Geotechnical Engineering Lab	1	PRACTICAL	5	2026-10-19 20:59:10
24CEL61	Environmental Engineering Laboratory	1	PRACTICAL	6	2026-10-19 20:59:10
24CEP71	Project Work Phase I	3	PRACTICAL	7	2026-10-19 20:59:10

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
```

enrollment_id	roll_no	course_code	semester	total_classes	attended_classes	is_condoned	created_at
1	24ITR001	24ITT31	3	60	49	0	2026-10-19 20:59:10
2	24ITR001	24ITC31	3	60	49	0	2026-10-19 20:59:10
3	24ITR001	24ITT32	3	60	51	0	2026-10-19 20:59:10
4	24ITR001	24ITC32	3	60	51	0	2026-10-19 20:59:10
5	24ITR001	24ITT33	3	60	53	0	2026-10-19 20:59:10
6	24ITR001	24MNT31	3	60	55	0	2026-10-19 20:59:10
7	24ITR001	24ITL31	3	30	26	0	2026-10-19 20:59:10
8	24ITR001	24ITL32	3	30	27	0	2026-10-19 20:59:10
9	24ITR001	24ITL33	3	30	28	0	2026-10-19 20:59:10
10	24ITR002	24ITT31	3	60	51	0	2026-10-19 20:59:10
11	24ITR002	24ITC31	3	60	51	0	2026-10-19 20:59:10
12	24ITR002	24ITT32	3	60	53	0	2026-10-19 20:59:10
13	24ITR002	24ITC32	3	60	52	0	2026-10-19 20:59:10
14	24ITR002	24ITT33	3	60	55	0	2026-10-19 20:59:10
15	24ITR002	24MNT31	3	60	56	0	2026-10-19 20:59:10
16	24ITR002	24ITL31	3	30	27	0	2026-10-19 20:59:10
17	24ITR002	24ITL32	3	30	28	0	2026-10-19 20:59:10
18	24ITR002	24ITL33	3	30	25	0	2026-10-19 20:59:10
19	24ITR003	24ITT31	3	60	53	0	2026-10-19 20:59:10
20	24ITR003	24ITC31	3	60	53	0	2026-10-19 20:59:10
21	24ITR003	24ITT32	3	60	55	0	2026-10-19 20:59:10
22	24ITR003	24ITC32	3	60	55	0	2026-10-19 20:59:10
23	24ITR003	24ITT33	3	60	56	0	2026-10-19 20:59:10
24	24ITR003	24MNT31	3	60	49	0	2026-10-19 20:59:10
25	24ITR003	24ITL31	3	30	28	0	2026-10-19 20:59:10
26	24ITR003	24ITL32	3	30	25	0	2026-10-19 20:59:10
27	24ITR003	24ITL33	3	30	26	0	2026-10-19 20:59:10
28	24CSR001	24CST31	3	60	49	0	2026-10-19 20:59:10
29	24CSR001	24CSC31	3	60	55	0	2026-10-19 20:59:10
30	24CSR001	24CST32	3	60	51	0	2026-10-19 20:59:10

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

LINK	https://github.com/thinarajkumar2008source/Student_Evaluation_Using_Java_JDBC